GYANASAI ADUSUMILLI

Jersey City, NJ | gadusumi@stevens.edu | linkedin.com/in/gyanasai | github.com/gyanasaiadusumilli

SUMMARY

Data Science master's student at Stevens Institute of Technology (graduating December 2026) who builds end-to-end machine learning systems in computer vision, remote sensing and retrieval-augmented generation (RAG). Hands-on with PyTorch, scikit-learn, OpenCV and LLM pipelines, from data collection and model training to deployed web demos. Seeking Machine Learning Engineer, AI Engineer and Data Scientist roles.

EDUCATION

Stevens Institute of Technology, Hoboken, NJ — Jan 2025 – Dec 2026 (expected)
Master of Science in Data Science | *GPA: 3.66/4.0*

SRM University AP, Amaravati, India — Sep 2020 – Jun 2024
Bachelor of Technology in Computer Science and Engineering, *Specialization in Artificial Intelligence*

Relevant coursework: Machine Learning, Artificial Intelligence, Probability and Statistics, Linear Algebra, Data Structures, Database Management, Parallel Computing, Software Engineering

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, Java, C

Machine Learning and AI: PyTorch, TensorFlow, Keras, scikit-learn, Deep Learning, Hugging Face, Sentence Transformers, FAISS, RAG, LLMs (Mistral-7B), NLP, Computer Vision, OpenCV, CNNs (ResNet, U-Net), Transfer Learning

Data and Analytics: pandas, NumPy, Matplotlib, Spark, Tableau, Power BI, MySQL, MongoDB, Oracle

Cloud and Tools: Git, GitHub, Streamlit, Google Earth Engine, Google Colab, AWS, Azure, Linux

PROFESSIONAL EXPERIENCE

OYO | *Technology Engineer Intern* — Jan 2024 – Jul 2024
Gurgaon, India

- Drove a **30% increase in user engagement** by launching a mobile-first application that gave users on-the-go access to critical system information.
- Partnered with the Ranking and Optimization team to analyze performance metrics for **500+ hotels**, identifying the factors behind higher conversion rates and average booking values.
- Wrote MySQL queries and built dashboards that turned operational data into actionable recommendations for project stakeholders.

PROJECTS

Water Body Detection from Satellite Imagery | *PyTorch*, *U-Net*, *scikit-learn*, *Earth Engine* — 2023; rebuilt 2026

- Built a data pipeline that pulls cloud-masked Sentinel-2 and Landsat-8 composites for 8 regions on 4 continents from Google Earth Engine, labeled with same-month JRC surface water data.
- Trained a U-Net CNN that reached **0.984 water IoU and 99.0% pixel accuracy** on Sentinel-2, about two-thirds fewer errors than the NDWI index baseline.
- Benchmarked SVM, random forest, gradient boosting, KNN and logistic regression against the CNN on 500,000 test pixels, using a spatial block split to prevent data leakage.

Colon Cancer Detection from Histopathology Images | *PyTorch*, *ResNet*, *OpenCV*, *Streamlit* — 2024; rebuilt 2026

- Fine-tuned an ImageNet-pretrained ResNet-18 with OpenCV CLAHE preprocessing to classify 10,000 H&E tissue images, reaching **100% accuracy and 1.0 ROC-AUC** on 1,500 held-out test images.
- Deployed a public **Streamlit web demo** that serves the model from Hugging Face Hub, with one-click reproducible training in Google Colab.
- Identified data leakage from augmented near-duplicates in the LC25000 dataset and documented a group-aware evaluation plan.

Symptoma: AI Symptom and Diagnosis Companion | *RAG*, *FAISS*, *LLMs*, *scikit-learn* — Jun 2025

- Co-built a four-stage pipeline combining ML classification, semantic search and LLM explanations to predict diseases from symptoms.
- Achieved **98% accuracy, precision, recall and F1** with a random forest classifier on structured symptom–disease data.
- Encoded free-text symptoms with Sentence Transformers (all-mpnet-base-v2) and indexed **240,000+ cases** in FAISS; Mistral-7B generated explanations grounded in the retrieved cases.

ACTIVITIES

- Competed in 5+ hackathons and was part of 2 winning teams building AI-driven environmental solutions.
- Paired 15 upperclassman mentors with incoming students, helping every participant connect with campus resources.